Assured PNT, Part 1

Navigation When GPS Cannot Be Trusted

Voice-track transcript for Michael L. Carroll

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Recording target: 17–18 minutes | 15 logical slides / 44 build pages

Recording notes

This script contains about 1,800 spoken words. The 17-minute-20-second timing plan assumes a deliberately measured technical delivery, including build cues and pauses; a brisk continuous read will be closer to 14–15 minutes. Text in the narration blocks is intended to be spoken. Production cues are not spoken. A short pause means about one second; a full pause means two to three seconds. Advance each build at the sentence indicated, leaving enough time for the viewer to read the new element.

Pronunciations: PNT is "pee-en-tee"; GNSS is "gee-en-ess-ess"; INS is "eye-en-ess"; IMU is "eye-em-you"; NIS is "en-eye-ess." Read P as "P," R as "R," and ν as "nu."

Slide 1 — Assured PNT

Target: 0:25

Production cue: Open on the complete title slide. No build.

> *What happens when a navigation system gives a precise, confident answer—and that answer is wrong? In this first episode on assured position, navigation, and timing, we will look at the system behavior required when GPS is blocked, jammed, or deliberately manipulated. The central theme is simple: resilience does not come from one perfect sensor. It comes from prediction, disciplined trust, and deliberate recovery.*

Slide 2 — A confident answer can still be wrong

Target: 1:15

Production cue: Begin on build 1. Advance once after "those are different failures."

> *The most dangerous navigation failure is not always a blank screen. It is a solution that still looks healthy after its reference has been manipulated. The receiver may continue producing position, velocity, and time. The output may be smooth. Its reported uncertainty may even remain small. Yet the answer can be moving away from truth.*
>
> *[Short pause.] A loss of availability says, "I no longer have the measurement." A loss of integrity says, "I have a measurement, but I should not rely on it." Those are different failures.*
>
> *[Advance.] That distinction gives us three rules for the rest of the episode. Availability is not integrity. Precision is not truth. And putting a measurement into a fusion algorithm does not*

automatically make that measurement trustworthy. A resilient system must keep its confidence tied to evidence.

Slide 3 — What PNT provides

Target: 1:05

Production cue: Reveal Position, Navigation, and Timing one at a time. Reveal the summary last.

[Build 1.] PNT begins with position: where am I?

[Build 2.] Navigation adds motion and intent: where am I going, how fast am I moving, and how is the vehicle oriented?

[Build 3.] Timing asks when an observation or event occurred. Timing is not an afterthought. It supports measurement alignment, communication, sensing, and the clock states inside a GNSS solution.

[Build 4.] The important engineering question is not whether every one of these quantities remains perfect under every condition. It is which combination of position, velocity, attitude, and time the mission actually needs—and what error it can tolerate. Assurance begins by turning that need into explicit performance limits.

Slide 4 — GPS is powerful—and physically fragile

Target: 1:20

Production cue: Start with the left block. Reveal the right block, then the concluding line.

GPS is an extraordinary navigation reference. It is global, common to many users, and capable of providing position, velocity, and precise time with very little infrastructure on the user platform. Its errors are also bounded over the long term, which makes it an excellent partner for inertial navigation.

[Advance.] But the signal arriving at the antenna is weak. Buildings, terrain, foliage, and vehicle structure can block it. Reflections can create multipath. Radio-frequency interference can overpower it. And a counterfeit signal can attempt to draw a receiver toward a false solution.

[Advance.] So our design objective cannot be nominal GPS accuracy alone. We have to design for both loss and manipulation. That means preserving useful navigation when measurements disappear, and protecting the estimator when measurements remain present but become misleading.

Slide 5 — Two failures, two different hazards

Target: 1:25

Production cue: Show jamming first. Reveal spoofing second. Reveal the Kalman-filter caveat last.

Jamming and spoofing create different estimator problems.

Jamming denies service. Interference raises the effective noise floor or overwhelms the authentic satellite signal. A receiver may lose tracking, lose satellites, or declare degraded performance. That is serious, but the failure is often observable: the external updates stop, and the navigator must bridge the gap.

[Advance.] Spoofing counterfeits the measurement. False GNSS-like data can appear plausible and can pull the estimated state away from reality. If the fusion system accepts that data repeatedly, it may also corrupt estimates of inertial sensor bias or clock error. The damage can therefore persist after the immediate event.

[Advance.] Statistical consistency checks can reject measurements that disagree with prediction. But a Kalman filter, by itself, does not authenticate a radio signal. A sophisticated false input can remain statistically plausible. Integrity requires multiple layers of evidence.

Slide 6 — Assurance is a behavior, not a sensor

Target: 1:20

Production cue: Advance through Protect, Detect, Respond, and Recover; then reveal the descriptions and final pill.

[Build 1.] Protect the system by reducing avoidable exposure: sound antenna installation, interference awareness, secure interfaces, and sensible operating procedures.

[Build 2.] Detect anomalies quickly. Monitor receiver health, signal conditions, estimator consistency, and agreement among independent sources.

[Build 3.] Respond in a controlled way. Isolate a suspect input, change its weight, switch modes, and continue only at a level of performance the system can support.

[Build 4.] Recover deliberately. A source that has returned is not necessarily a source that should be trusted immediately.

[Build 5.] These four actions connect technical design to operations: prevent what we can, recognize what remains, degrade gracefully, and test sources before re-admission.

[Build 6.] The system must know what it trusts—and why. That is behavior of the complete architecture, not a property we can buy in a single sensor.

Slide 7 — Inertial navigation provides the bridge

Target: 1:20

Production cue: Begin with INS strengths. Reveal the uncertainty plot and limitation together.

An inertial navigation system is the natural bridge through a GPS outage. Once initialized, it is self-contained. Accelerometers and gyroscopes support a high-rate estimate of position, velocity, and attitude. Because the measurement is mechanical rather than radio-frequency based, an IMU is not disabled by an RF jammer. And over short intervals, inertial propagation is smooth and responsive.

[Advance.] The price is drift. Accelerometer bias is integrated into velocity and then position. Gyroscope bias creates attitude error, which projects gravity and specific force in the wrong direction. Noise and imperfect models accumulate.

The curve is conceptual, not a universal error law. Its message is operational: uncertainty grows with outage time. The vehicle may continue only until the predicted uncertainty reaches the mission limit. INS gives us time, but it does not give us infinite time.

Slide 8 — Resilience comes from complementary errors

Target: 1:15

Production cue: Show GNSS plus INS, add the other aids, then reveal the design test.

[Build 1.] GPS and INS work well together because their errors are complementary. Inertial sensors provide continuity and rapid dynamics but drift. GNSS provides an external reference that can bound that drift, but its signals can disappear or be deceived.

[Build 2.] The same reasoning extends to air data, barometric altitude, terrestrial radio ranging, terrain matching, and vision. Each source observes some part of the state, with its own geometry, update rate, environment, and failure mechanisms.

[Build 3.] Simply adding sensors is not resilience. A useful source has failure modes that are sufficiently independent of the sources already present. Its measurements must also make the mission-critical states observable. Ten sensors that share the same hidden dependency may provide less assurance than two carefully chosen sources based on different physics.

Slide 9 — The Kalman filter is the integration engine

Target: 1:30

Production cue: The full diagram is visible. Trace it from left to right with the narration.

The Kalman filter gives this architecture a disciplined predict-and-update cycle.

IMU samples drive the prediction. The mechanization propagates the navigation state, while the process model propagates the covariance, P. The superscript minus means this is the prior estimate: our best prediction before using the new external measurement.

External measurements then pass through validation. Does the observation agree with what the predicted state and its uncertainty say is plausible? Do other independent sources support it? Is the receiver healthy? If the evidence is adequate, the update corrects the state and reduces uncertainty. The superscript plus marks the posterior estimate.

If the evidence is poor, the system can reject the measurement, isolate the source, or increase its assumed noise so that it receives less weight. The important point is that prediction carries the solution through a gap, while trusted updates bound drift. An update is beneficial only when accepting it is safer than continuing without it.

Slide 10 — A practical error-state model

Target: 1:10

Production cue: Reveal the state vector, then the table, then the design-choice pill.

[Build 1.] Many integrated navigators use an error-state filter. The inertial mechanization carries the full navigation solution, and the Kalman filter estimates small errors around that solution.

[Build 2.] Here the state contains position and velocity error, small attitude error, accelerometer and gyro biases, and receiver clock bias and drift. Estimating the inertial biases is especially important: an external reference does more than correct the current position. It calibrates error sources that would otherwise drive future drift.

[Build 3.] This vector is an example, not a universal recipe. The state is a design choice. Add a quantity when the mission needs it, when a credible model exists, and when the available measurements can actually observe it. A larger state that cannot be supported by the measurement geometry can create false confidence rather than capability.

Slide 11 — Trust begins with the innovation

Target: 1:30

Production cue: Reveal the innovation, then NIS, then the validation list, and finally the caveat.

[Build 1.] The innovation is the measured value, z, minus the value predicted from the prior state. In plain language: what did the sensor say that the filter did not expect?

A large difference may indicate interference, a failed sensor, a bad model, or an aggressive maneuver the estimator did not represent well. The residual is evidence, not yet a diagnosis.

[Build 2.] The normalized innovation squared—the NIS—scales the residual by its predicted covariance, S. This matters because ten meters may be alarming in one geometry and entirely plausible in another. Under the filter's modeling assumptions, the NIS can be compared with a statistical threshold.

[Build 3.] A practical validator does more. It checks persistence over time, compares independent sources, considers signal and receiver health, and uses explicit recovery logic.

[Build 4.] A gate tells us that a measurement is inconsistent with the current model and evidence. It does not prove whether the cause was spoofing, jamming, multipath, a sensor fault, or a bad assumption. Good assurance preserves that distinction.

Slide 12 — What the system should do during an outage

Target: 1:25

Production cue: Advance through the timeline phases, then reveal the graceful-degradation block.

[Build 1.] Before an event, trusted GNSS updates keep the inertial solution and its bias estimates calibrated.

[Build 2.] When an anomaly appears, the system should detect it promptly and stop treating the affected input as ordinary truth. That decision may be based on several monitors rather than one threshold crossing.

[Build 3.] During the outage, the INS propagates the state. Other trusted aids can continue updating the filter. Covariance, P, must grow consistently with the remaining sensor errors and the weaker measurement geometry.

[Build 4.] When the source returns, validate it before reuse. Compare it against the propagated solution and independent aids. Require a suitable interval of consistent behavior. Re-admission may be gradual rather than instantaneous.

[Build 5.] This is graceful degradation: continue only while estimated uncertainty remains inside a declared operational bound. When it does not, transition to a safe mode. The estimate should never look healthier merely because its best reference disappeared.

Slide 13 — Four design rules for assured PNT

Target: 1:15

Production cue: Reveal one numbered rule at a time, then reveal the final sentence.

[Build 1.] First, diversify. Combine sources that use different physics and do not share all of the same failure modes.

[Build 2.] Second, manage trust. A precise measurement is not necessarily an authentic measurement. Weight must follow evidence.

[Build 3.] Third, expose uncertainty. Propagate covariance realistically, validate it where possible, and compare it with the error the mission can tolerate.

[Build 4.] Fourth, plan recovery before the failure occurs. Define how a source is declared suspect, how it is isolated, what conditions permit continued operation, and how it is tested before re-admission.

[Build 5.] These rules reinforce one another. Assurance is the controlled behavior of the whole navigation system—including its estimator, monitors, modes, interfaces, and operating procedures.

Slide 14 — The central idea

Target: 0:55

Production cue: Hold on the complete summary slide.

The central idea can be reduced to three verbs: predict continuously, challenge measurements, and recover deliberately.

A resilient navigator does not promise perfect availability. No physical system can. It promises that performance is known, uncertainty is visible, and behavior remains safe when important inputs are lost or challenged.

In Part Two, we will move from architecture to aiding sources. We will compare barometric, radio, terrain, and vision-based measurements, and then examine how terrain-referenced navigation can bound inertial drift when GNSS is not available.

Slide 15 — Primary references

Target: 0:10

Production cue: Leave the references on screen for at least ten seconds. Do not narrate the individual URLs.

The primary government references used for this episode are shown here and are also linked in the video description.

Source links for the video description

- FAA, *GPS and GNSS Interference Resource Guide*, 2025:
  FAA GPS/GNSS Interference Resource Guide page
- GPS.gov, *Resilience Through Responsible Use of PNT*:
  https://www.gps.gov/resilience-through-responsible-use-pnt
- NIST IR 8323 Rev. 1, *Foundational PNT Profile*, 2023:
  https://doi.org/10.6028/NIST.IR.8323r1
- DHS, *Resilient PNT Reference Architecture*, 2022:
  https://www.dhs.gov/sites/default/files/2022-06/22_0609_st_resilient_pnt_ra.pdf
- U.S. DOT, *Complementary PNT Action Plan*, 2024:
  https://www.transportation.gov/pnt/complementary-pnt-action-plan
- DHS, *Best Practices for Resilient PNT Supporting Critical Infrastructure*, 2025:
  https://www.dhs.gov/sites/default/files/2025-02/25_0220_st_pnt_best_practices_ci.pdf